Juliette Bruce's Scholarship, Teaching, Service

a single conviction runs through much of my research, teaching, and service: a desire to help people explore and contribute to mathematics by making hidden structures visible.

Scholarship. My research is in the intersection of algebraic geometry and commutative algebra, which both focus on studying the duality between systems of polynomial equations and the geometry of their solution sets. As described i my future research plan, my work follows three connected threads.

Syzygies My doctoral work studied the relations among systems of polynomial equations, often called syzygies, that describe higher dimensional shapes. The equations defining a shape are rarely independent, but instead often statisfy relations, and the relations satisfy further relations, and so on. Homological algebra provides us tools to organize this cascade of relations, and understand what it tells us about the geometry of the underly shapes. In my work on asymptotic syzyges, I studied how these syzygies change when we descibe the same shape using increasingly complicated coordinate. THis work resulted in publications in the *Journal of Algebra* ADD STUFF the *Journal of Commutative Algebra* ADD STUDD , and *Experimental mathematics* ADDD Together this work has seen BLAHD

Regularity Regularity is a measure of complexity that connects the geometry of a shape with the BEDEDED. Classically, it is a single number, however, together with Lauren Cranton Heller and Mahrud Sayrafi, I have studied how in behaves BLAH BLAH BLAH. This work has resulted in one publication in *Advances in Mathematics* and two pre-prints under-review at journals or roughly the same quality.

Tropical Geometry of Moduli Spaces

This has resulted in one publication in *Geometry & Topology*, and three-preprints under review.

Dissemination Since joining Dartmouth in Fall 2024 I have posted five new pre-prints, all of which are under-review at peer-reviewed journals. My paper characterizing multigraded regularity was accepted in *Advances in Mathematics*. Since June 2024 I have given 12 invited research talks at seminars, conferences, and conferences. have attended six invitation-only research workshops since 2024 including two at Oberwolfach, the Banff International Research Station, the Fields Institute, and the American Institute of Mathematics.

Teaching and Mentoring. My first experiences teaching were as an undergraduate course assistant in inquiry-based courses at the University ofM Michigan. In these courses the students worked together to build and discover mathematics for themselves through carefully curated mix of group work, lecture, and carefully chosen problems. That experience still shapes much of my teaching. I want students to have opportunities to discover and work out an idea, explain it to someone else, and develop their own questions

At Dartmouth, I have taught introductory calculus, undergraduate abstract algebra, and an array of graduate courses. At the undergraduate level most of my teaching focus has been on refining and improving Math 8: Calculus of One and Several Variables. This course serves as the first mathematics course many Dartmouth students take, and at least from the students in my first class, has a somewhat intimidating reputation. I have spent significant time making BLAH BLAH BLAH a point of support for students. While maintaining exceptionally high standards, BALH BLAH BLHA.

The advanced courses I have taught (Math 101, 121, 125) are where my teaching and research meet most directly. Math 125 was a topics course using toric varieties as a touchstone to provide students with grounded introduction to algebraic geometry. Math 121 was a commutative algebra course, which I designed around a series of sixteen inquiry-based worksheets, that build on work of Karen Smith. These worksheets are available freely on my website, and BLA BLAH BLAH. Additionally, i have taught Math 150, a gradaute reading course, two times; in Fall 2026 I led a group of 10 students FINISH, and in Winter 2025 I led a group of 15 undergradaute and graduate students learning about one of the jewels of 20th century mathematics: The Weil Conjectures.

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Mentoring students is one of the parts of my job I find most rewarding. Currently I m advising two Ph.D. students: Will DeGroot (4th year) and Jacob Lehmann Duke (3rd year). Their projects connect with my own interests in computation and the complexity of equations on toric varieties. I expect that within the year I will become Alex Moon's (2nd year) Ph.D. advisor as I have been helping them prepare for their advancement exam, and advising them on a small research project. Since Summer 2026 I have mentored Amy Q Li, a new post doc in the mathematics department.

Beyond Dartmouth, in summer 2026 I led three undergraduates on a summer search program through *Count Me In* held at the University of Edinburgh. All three of these students are now in the process of applying to graduate school. In Fall 2026 I am one of two lectures for the Preliminary Arizona Winter Schoo, a nine-week online course introducing approximately 300 advanced undergraduate students and early graduate students from around the world to recent advancements in computational algebraic geometry.

Service and Community. Mathematics is a deeply collaborative field, and much of its culture is passed along informally.

At Dartmouth. In fall 2025 I founded *Unwritten Rules*, a graduate seminar that seeks to make the hidden curriculum and expectation of an academic career explicitly. Since its founding ADD THINGS WE MEET 1-2 TIMES A MONTH (CONFERENES, FUNDING. JOURNALS). Since 2025 I serve as the faculty co-advisor for Dartmouth's student chapter of the Association for Women in Mathematics. DINNER ON GOING TO GRAD SCHOOL. In November 2024 I organized the Algebraic Geometry Northeastern Series (AGNES) and BATMOBILE conferences at Dartmouth, which brought over 150 mathematicians to Dartmouth to discuss recent breakthroughs in algebraic geometry. Currently, I am in the early planning stages of organizing *GEMS of Commutative Algebra*, a conference building research networks for early-career researchers in commutative algebra, which I hope to held at Dartmouth in 2028.

In the profession. I was the inaugural president for *Spectra, the Association for LGBTQ+ Mathematicians*, and led the adoption of bylaws,the creation of a invited lecture series at the Joint Mathematics Meetings, and fundraised over $20,000 in seed funding. I am currently serving on the board of trustees for *Spectra* working to guide the organization to longterm ADD PUFF. From 2023 to 2025 I served on SLMath's Committee on Women in Mathematics. Currently I serve on the American Mathematical Society's Committee on Engagement, Participation, and Advancement and the Committee on AMS-Simons Travel grants. I have reviewed grants for the National Science Foundation, refereed for more than a dozen different journals. In my career I have organized 18 conferences and workshops that have brought together over 1000 mathematicians. With co-organzier, I have submitted a proposal to the Fields Institute to organize a thematic semester in combinatorial algebraic geometry.